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THE ROYAL SOCIETY

# Oblique impact with friction and tangential compliance

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The present temporal analysis of impact between two rough and compliant hard bodies employs a lumped-parameter model of the small region of contact that suffers significant deformations. This model represents effects of both normal and tangential compliance by means of discrete elements. By employing specific compliance relations one obtains *smooth dynamics* of the time-dependent processes of stick and sliding that take place in the contact region during an oblique collision. Effects of compliance and friction on the contact process in an eccentric impact configuration are obtained for planar changes in relative motion and for energy dissipation during impact. The solutions are compared with results obtained with the assumption of negligible compliance in order to determine the range of impact velocities and structural parameters wherein tangential compliance significantly affects the results.

**Keywords: collision; impact; friction; contact compliance; stick–slip process**

## 1. Introduction

Analyses of the dynamics of low-speed or rigid-body collisions between rough bodies in non-collinear configurations have shown that at the contact point slip reversal will occur if the angle of incidence is small and there is significant friction. When the angle of incidence is somewhat larger, the contact process changes to slip–stick. At sufficiently large angles of incidence, gross or continuous slip occurs throughout the contact period.

In order to achieve a numerically more efficient calculation of the dynamics of collision, some authors employ non-smooth dynamics to obtain equations of motion in terms of an independent variable: the *normal impulse* $p$ at the contact point. Tangential impulse is then related to the normal impulse through a coefficient: the terminal impulse ratio (see, for example, Brach 1991; Pfeiffer & Glocker 1996; Chatterjee & Ruina 1998). While the ratio of tangential to normal impulse equals the coefficient of friction during continuous sliding in any direction, if the direction of sliding varies, then some approximation is required in order to obtain the respective components of tangential impulse at termination of collision. Clever means have been devised to relate the ratio of tangential to normal impulse to the angle of incidence at the contact point (see, for example, Smith 1991); nevertheless, this coefficient is an approximation that represents an average force over the contact period. Can such a coefficient give the correct dependence on the angle of incidence, the impact configuration and the coefficient of restitution?

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The question is irrelevant if the angle of incidence is large enough so that there is continuous slip during contact. If friction is large and the angle of incidence is small, however, initial sliding is brought to a halt before the termination of contact. In this case the impulse ratio depends on the angle of incidence at the contact point as well as the impact configuration and any sources of energy loss in collision (Stronge 1994). Because these interactions are complex, the present investigation abandons non-smooth dynamics that is dependent on finite changes of impulse at the contact point; instead we develop an explicitly time-dependent approach to calculating velocity changes during impact. This approach is independent of the specific compliance of the contact region however.

For collinear collisions, Maw *et al*. (1976, 1981) investigated effects of tangential compliance on the velocity changes occurring during impact between bodies with spherical contact surfaces. They obtained an elasticity solution for the traction distribution in a circular contact patch between rough collinear solid bodies where the bodies are subjected to both normal and tangential relative displacement. They used the resulting force–displacement relation to calculate velocity changes as a function of *time* in a colliding sphere with small compliance and showed that during a collision there is a reversal of the direction of slip if the angle of incidence is small. For collinear collisions, this effect cannot happen if tangential compliance is negligible (in comparison with the normal compliance). Time-dependent impact analyses that employed single-degree-of-freedom lumped-parameter representations of the normal compliance were used by Cundall & Strack (1979) and Mills & Nguyen (1992). Later, Stronge (1994) developed a two-degrees-of-freedom extension of this representation to show that, for collinear configurations, the continuum elastic impact response could be simulated by a lumped-parameter model employing separate normal and tangential compliances. For the problem of an elastic disc colliding with a rough half-space, calculations using Stronge's lumped-parameter model gave results for the variation of relative velocity at separation as a function of angle of incidence that were almost identical to the experimental data of Maw *et al*. (1981). Those solutions exhibited reversal in the direction of relative tangential velocity for small angles of incidence.

## 2. Analytical model

### (*a*) *Rationale for impact model*

Impact occurs when two bodies come together at a contact point with some relative velocity in the direction normal to the common tangent plane. In order to prevent interpenetration, hard bodies undergo deformation in a small region localized around the initial point of contact. With small compliance of this contact region, the small surface displacements that result from these deformations are sufficient to develop the large contact forces which drive the bodies apart during impact. These large forces quickly halt the relative approach of the contact surfaces and subsequently drive the bodies apart. The period of approach for the normal component of relative velocity is termed *compression*, while the subsequent period of contact decompression is termed *restitution*. For hard bodies the period of collision (i.e. the sum of the compression and restitution periods) is very small; e.g. impact between a golf ball and a club head takes place in less than 1 ms (Johnson & Liebermann 1996). During this brief period of collision there are large changes in the relative velocities but the associated

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relative displacements are negligibly small. Consequently, we can assume that the configuration and hence the inertia properties are invariant during impact.

In order to derive equations for the changes in relative velocity at the contact point in a collision between hard bodies, we represent the small deforming region by a deformable particle and consider the remainder of the system to consist essentially of rigid bodies. With this artifice, the contact forces acting on the rigid bodies during collision are equal in magnitude but opposite in direction. This analytical model is representative of collisions between hard bodies where the contact area is small in comparison with characteristic dimensions of the bodies and the deformation is localized around the point of initial contact. During impact it results in *smooth* (i.e. continuous) variations of the relative velocity at the contact point as a function of either time or the normal component of impulse.

The effect of tangential compliance on processes of contact sliding and stick requires consideration of variations in relative displacements at the contact point in addition to the changes in relative velocity. To be consistent with rigid-body modelling, these displacements must be sufficiently small so that they do not affect inertia properties of the bodies, i.e. this requires that contact compliance is small. In the present model it is not the magnitude of the compliance, but rather the ratio of tangential to normal compliance, that significantly affects contact processes during collision.

### (*b*) *Dynamics of rigid-body collision*

Consider two colliding bodies B and B′ which coincide at a single contact point C during a collision. The colliding bodies have masses $M$, $M'$ and contact points C, C′. Suppose at least one of the bodies has a smooth contour so that there is a common tangent plane (CTP) that passes through C. Define unit vectors $\boldsymbol{n}_1$ and $\boldsymbol{n}_3$ that are parallel and normal to the CTP, respectively. In order to focus attention on effects of tangential compliance, the present paper considers the case of planar collision, where the velocity changes occur in the $\boldsymbol{n}_1, \boldsymbol{n}_3$-plane only. The contact points first touch at an instant of time $t = 0$, termed incidence, when the bodies come together with some initial relative velocity $\boldsymbol{v}(0) \equiv \boldsymbol{V}_{\mathrm{C}}(0) - \boldsymbol{V}'_{\mathrm{C}'}(0)$. The normal component of the incident relative velocity is negative, $\boldsymbol{v}(0) \cdot \boldsymbol{n}_3 = v_3(0) < 0$ and it remains negative throughout the period of compression. At C and C′ there are equal but opposite contact forces $\boldsymbol{F}$ and $\boldsymbol{F}'$ that act on the bodies and drive them apart, $\boldsymbol{F} = (F_1, F_3)$.

We assume that the system configuration is invariant during collision, i.e. in the reference frame of the CTP the inertia properties of the colliding bodies are constants. For planar problems, inertia properties for each body can be specified in terms of the radii of gyration $k$ and $k'$ about the respective centres of mass and the position vectors $\boldsymbol{r} = (r_1, r_3)$ and $\boldsymbol{r}' = (r'_1, r'_3)$ from the respective centres of mass to the contact point C.

The relative accelerations at the contact point $\mathrm{d}\boldsymbol{v}(t)/\mathrm{d}t = (\mathrm{d}v_1/\mathrm{d}t, \mathrm{d}v_3/\mathrm{d}t)$ can be expressed in terms of these variables (Stronge 1993):

$$\begin{Bmatrix} \mathrm{d}v_1/\mathrm{d}t \\ \mathrm{d}v_3/\mathrm{d}t \end{Bmatrix} = m^{-1} \begin{bmatrix} \beta_1 & -\beta_2 \\ -\beta_2 & \beta_3 \end{bmatrix} \begin{Bmatrix} F_1 \\ F_3 \end{Bmatrix}, \qquad m^{-1} \equiv M^{-1} + M'^{-1}. \tag{2.1}$$

This expression is valid for initial conditions that include rotational as well as translational velocities of the bodies. Elements of the inverse of inertia matrix are the

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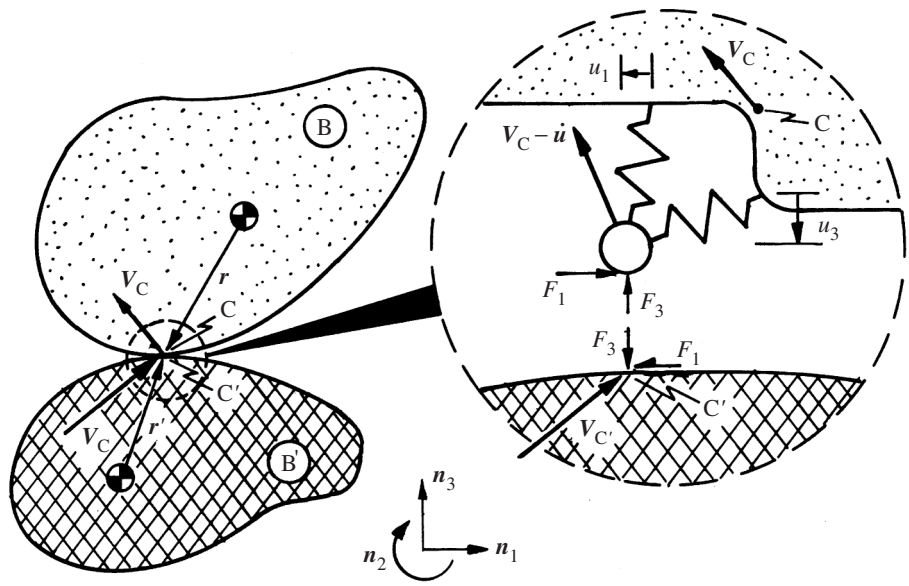


Figure 1. Analytical model for planar impact between two hard bodies with separate normal and tangential compliance at the contact point C.

inertia coefficients $\beta_i$, defined as

$$\beta_1 \equiv 1 + \frac{mr_3^2}{Mk^2} + \frac{mr_3'^2}{M'k'^2}, \qquad \beta_2 \equiv \frac{mr_1 r_3}{Mk^2} + \frac{mr_1' r_3'}{M'k'^2}, \qquad \beta_3 \equiv 1 + \frac{mr_1^2}{Mk^2} + \frac{mr_1'^2}{M'k'^2},$$

where $\beta_1 > 1$, $\beta_3 > 1$ and $\beta_1\beta_3 > \beta_2^2$.

### (c) *Contact compliance*

In this lumped-parameter model, the compliance of a small elastoplastic region around the contact point is represented by discrete elements oriented in the normal and tangential directions. The elements are connected to a *massless particle* which can either stick or slide on the surface of the second body at the contact point C′ as shown in figure 1. For bilinear compliance that incorporates hysteresis of the normal compliant element only, the contact forces $F_i$ are related to the small extensions $u_i$ of compliant elements during separate periods of compression $0 \leqslant t \leqslant t_c$ and restitution $t_c < t < t_f$

$$\begin{Bmatrix} F_1 \\ F_3 \end{Bmatrix} = -\kappa \begin{bmatrix} \eta^{-2} & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_3 \end{Bmatrix}, \qquad \dot{u}_3 < 0 \quad \text{(compression)}, \tag{2.2 a}$$

$$\begin{Bmatrix} F_1 \\ F_3 \end{Bmatrix} = -\kappa \begin{bmatrix} \eta^{-2} & 0 \\ 0 & e_*^{-2} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_3 \end{Bmatrix} + \kappa \begin{Bmatrix} 0 \\ (e_*^{-2} - 1)u_{3c} \end{Bmatrix}, \quad \dot{u}_3 > 0 \quad \text{(restitution)}, \tag{2.2 b}$$

where $\eta^{-2}$ is the ratio of tangential to normal stiffness, $t_c$ is the time of transition from compression to restitution $\dot{u}_3(t_c) = 0$, and $u_{3c} = -\kappa^{-1}F_3(t_c)$ is the maximum normal

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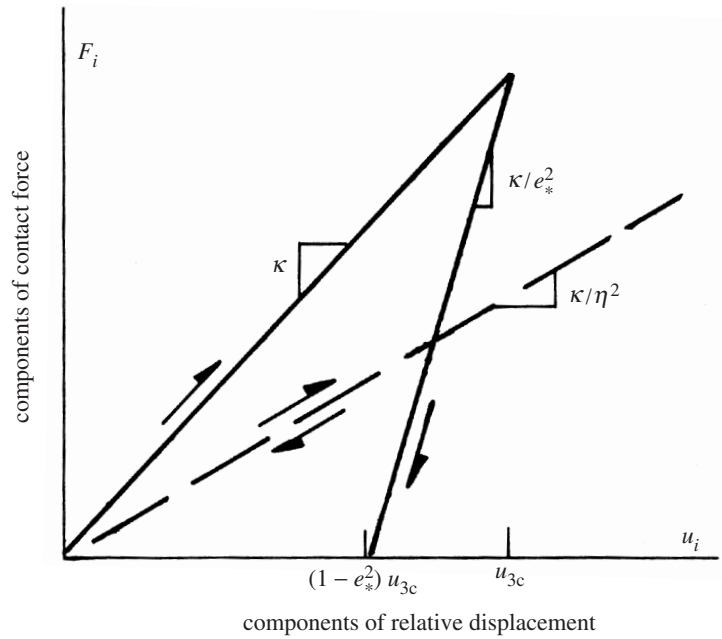


Figure 2. Force–deflection relations for normal and tangential elements at contact point C. The normal force $F_3$ is elastic-plastic, while the tangential force $F_1$ is elastic; the elements have elastic stiffnesses that differ by $\eta^2$.

compression.† For normal force these relations give a loading–unloading hysteresis loop which equates to the energetic coefficient of restitution $e_*$; also they provide continuity of force and displacement at maximum indentation $u_{3c}$ (Stronge 1990). The hysteresis of normal force results in a terminal indentation $u_3(t_f) = (1-e_*^2)u_{3c}$. The tangential force–displacement relation is assumed to be elastic. These normal and tangential stiffnesses are illustrated in figure 2, where it is assumed that $v_3(0) < 0$. They are appropriate for bodies composed of elastoplastic materials that collide at moderate speeds which result in only imperceptible final indentation (i.e. the small final indentation results from sub-surface plastic deformation contained beneath the contact surface (Stronge 2000)).

### (*d*) *Friction during sliding of rough contacts*

Dry friction due to sliding of rough contacting surfaces can be represented by the Amontons–Coulomb law of dry friction (Johnson 1985). While the contact is sliding, this law relates the normal and tangential components of contact force by a *coefficient of limiting friction* $\mu$.‡ With tangential compliance, the direction of sliding $\hat{s}(t)$ is defined as $\hat{s} \equiv \mathrm{sgn}(v_1 - \dot{u}_1)$, where $\dot{u}_1(t)$ is the rate of extension of the tangential compliant element and $v_1(t)$ is the relative velocity across the deformable region

† For elastic bodies with Poisson's ratio $\nu = 0.3$, a spherical contact surface on a half-space gives $\eta^2 = 1.21$ (see Johnson 1985). This value has been used in the examples of this paper except where noted.

‡ For clarity in demonstrating effects of friction, the static and dynamic coefficients of friction are assumed to be equal.

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around the contact point:

$$\left.\begin{array}{lll} \text{stick:} & \hat{s}=0, & |F_1|/F_3<\mu, \\ \text{sliding:} & \hat{s}=\pm 1, & F_1/F_3=-\mu\hat{s}. \end{array}\right\} \tag{2.3}$$

### (e) *Rate of work (energy dissipation) during collision*

The rate of work $\dot{W}$ done by contact forces on the colliding body is given by

$$\dot{W}=\boldsymbol{F}\cdot\boldsymbol{v}, \tag{2.4}$$

while the rate of energy dissipation by friction $\dot{D}_1=F_1(v_1-\dot{u}_1)$, where $(v_1-\dot{u}_1)$ is the speed of sliding. The total rate of work is the sum of tangential and normal components $\dot{W}=\dot{W}_1+\dot{W}_3$. The rate of work done on the colliding bodies by the tangential force $\dot{W}_1$ differs from the rate of dissipation because of the strain energy absorbed by the tangential compliant element. In the present model the tangential element is elastic, so that finally the strain energy stored in the tangential element vanishes at separation $t_{\mathrm{f}}$, i.e. $W_1(t_{\mathrm{f}})=D_1(t_{\mathrm{f}})$.

## 3. Solution for initial stick

### (a) *Equation of motion for stick*

During stick the sliding velocity vanishes $0=v_1-\dot{u}_1$ so that both compliant elements have a rate of extension equal to the relative velocity across the respective element, $\dot{u}_k=v_k$, $k=1,3$. Also, the contact force depends on relative extension of compliant elements in accord with equation (2.2). Hence, after substituting these expressions into (2.1), we obtain the equations of motion for the period of compression,

$$\begin{Bmatrix} \mathrm{d}^2u_1/\mathrm{d}t^2 \\ \mathrm{d}^2u_3/\mathrm{d}t^2 \end{Bmatrix}+\omega_0^2\begin{bmatrix} \beta_1\eta^{-2} & -\beta_2 \\ -\beta_2\eta^{-2} & \beta_3 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_3 \end{Bmatrix}=\begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \quad \omega_0^2\equiv\frac{\kappa}{m}, \quad \omega_0 t\leqslant\tfrac{1}{2}\pi, \tag{3.1 a}$$

and, for the subsequent period of restitution beginning at $\omega_0 t=\pi/2$, equations (2.1) and (2.2 *b*) give

$$\begin{Bmatrix} \mathrm{d}^2u_1/\mathrm{d}t^2 \\ \mathrm{d}^2u_3/\mathrm{d}t^2 \end{Bmatrix}+\omega_0^2\begin{bmatrix} \beta_1\eta^{-2} & -\beta_2 e_*^{-2} \\ -\beta_2\eta^{-2} & \beta_3 e_*^{-2} \end{bmatrix}\begin{Bmatrix} u_1 \\ u_3 \end{Bmatrix}=\omega_0^2\begin{Bmatrix} -\beta_2(e_*^{-2}-1)u_{3\mathrm{c}} \\ \beta_3(e_*^{-2}-1)u_{3\mathrm{c}} \end{Bmatrix}, \quad \omega_0 t>\tfrac{1}{2}\pi, \tag{3.1 b}$$

where displacement components are continuous at the transition from compression to restitution. For the period of compression this equation of motion gives modal frequencies (eigenvalues),

$$\frac{\omega_i^2}{\omega_0^2}=\frac{\beta_1\eta^{-2}+\beta_3}{2}\left\{1\mp\sqrt{1-\frac{4\eta^{-2}(\beta_1\beta_3-\beta_2^2)}{(\beta_1\eta^{-2}+\beta_3)^2}}\right\}, \qquad i=1,2, \tag{3.2 a}$$

while for the period of restitution the modal frequencies are

$$\frac{\omega_i^2}{\omega_0^2}=\frac{\beta_1\eta^{-2}+\beta_3 e_*^{-2}}{2}\left\{1\mp\sqrt{1-\frac{4\eta^{-2}e_*^{-2}(\beta_1\beta_3-\beta_2^2)}{(\beta_1\eta^{-2}+\beta_3 e_*^{-2})^2}}\right\}. \tag{3.2 b}$$

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### (*b*) *Relative displacement for initial stick*

If the relative velocity is such that the collision begins with an initial period of stick, then the equations of motion (3.1 *a*) with initial conditions $u_1(0) = u_3(0) = 0$ have a solution

$$\left.\begin{aligned} u_1 &= A_1 \sin\omega_1 t + A_2 \sin\omega_2 t, \\ u_3 &= B_1 \sin\omega_1 t + B_2 \sin\omega_2 t. \end{aligned}\right\} \tag{3.3}$$

The additional conditions for initial stick, $\dot{u}_k(0) = v_k(0)$ give the coefficients for the modal solution

$$\frac{A_i\omega_i}{v_3(0)} = (-1)^i \frac{1 + (B_i/A_i)v_1(0)/v_3(0)}{(B_2/A_2) - (B_1/A_1)}, \qquad \frac{B_i}{A_i} = \frac{1}{\beta_2}\left(\frac{\beta_1}{\eta^2} - \frac{\omega_i^2}{\omega_0^2}\right), \qquad i = 1, 2. \tag{3.4}$$

### (*c*) *Initial stick or slip?*

At an early time $t = 0 + \varepsilon$, where $\varepsilon \ll \omega_0^{-1}\pi/2$, the solution (3.3), (3.4) gives a contact force with components

$$\left.\begin{aligned} F_1(\varepsilon) &= -\kappa\eta^{-2} v_1(0)\varepsilon, \\ F_3(\varepsilon) &= -\kappa v_3(0)\varepsilon. \end{aligned}\right\} \tag{3.5}$$

Thus, for Amontons–Coulomb friction, there is initial stick at the contact point C if the angle of incidence $\psi_0 = \tan^{-1}|v_1(0)/v_3(0)|$ is small; namely

$$\left|\frac{v_1(0)}{v_3(0)}\right| < \mu\eta^2. \tag{3.6}$$

Because this limit is independent of inertia properties of the colliding bodies, the largest angle of incidence for initial stick of non-collinear collisions is identical to that obtained for collinear collisions (Stronge 1994).

## 4. Solution for gross slip

During planar collisions there is continuous slip in the initial direction (gross slip) unless the initial speed of sliding is relatively small.

### (*a*) *Equation of motion for gross slip*

Irrespective of slip or stick, during contact the normal component of rate of displacement $\dot{u}_3 = v_3$. If the contact is sliding, the tangential force is related to the normal force by the law of friction. Similarly, the tangential displacement is related to the normal displacement by friction and the ratio of compliances, $\eta^2$. For sliding during compression, equations (2.1)–(2.3) give

$$\left.\begin{aligned} u_1 &= -\mu\hat{s}\eta^2 u_3, \\ \frac{\mathrm{d}^2 u_3}{\mathrm{d}t^2} + \omega_0^2\beta_{32}u_3 &= 0, \end{aligned}\right\} \tag{4.1 a}$$

where $\beta_{32} \equiv (\beta_3 + \mu\hat{s}\beta_2)$. For a subsequent period of restitution, the equations of motion are

$$\left.\begin{aligned} u_1 &= -\mu\hat{s}\eta^2[e_*^{-2}u_3 - (e_*^{-2} - 1)u_{3\mathrm{c}}], \\ \frac{\mathrm{d}^2 u_3}{\mathrm{d}t^2} + \omega_0^2\beta_{32}e_*^{-2}u_3 &= \omega_0^2\beta_{32}(e_*^{-2} - 1)u_{3\mathrm{c}}. \end{aligned}\right\} \tag{4.1 b}$$

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If the contact point is sliding, equations (4.1 *a*) and (4.1 *b*) apply rather than equations (3.1 *a*) and (3.1 *b*).

### (*b*) *Relative displacement at the contact point*

The ordinary differential equations (3.1) can be solved to obtain the relative displacement $u_i(t)$ across the deformable particle at the contact point C. For impact between solids with no initial pre-stress, the initial conditions are $u_i(0) = 0$, $\dot{u}_3(0) = v_3(0) = -v_0$; hence, for gross slip during compression we obtain

$$\left.\begin{aligned} u_1(t) &= \frac{\mu\hat{s}\eta^2 v_0}{\beta_{32}^{1/2}\omega_0}\sin(\beta_{32}^{1/2}\omega_0 t),\\ u_3(t) &= \frac{-v_0}{\beta_{32}^{1/2}\omega_0}\sin(\beta_{32}^{1/2}\omega_0 t),\end{aligned}\right\}\quad 0 < t \leqslant t_{\mathrm{c}}. \tag{4.2 a}$$

Maximum compression $u_{3\mathrm{c}} \equiv u_3(t_{\mathrm{c}})$ occurs at time $t_{\mathrm{c}} \equiv \pi/(2\beta_{32}^{1/2}\omega_0)$ when the relative displacement equals $u_{3\mathrm{c}} = -v_0/\beta_{32}^{1/2}\omega_0$. Thereafter the displacement magnitude decreases during restitution. For the bilinear compliance relation (2.2 *b*) and continuity of both displacements and time at the transition from compression to restitution, the components of relative displacement during restitution are obtained as

$$\left.\begin{aligned} u_1(t) &= -\mu\hat{s}\eta^2 u_{3\mathrm{c}}\left[1 - e_*^2 + e_*^2\sin\left(\frac{\beta_{32}^{1/2}\omega_0}{e_*}(t - t_{\mathrm{c}}(1-e_*))\right)\right],\\ u_3(t) &= u_{3\mathrm{c}}\left[1 - e_*^2 + e_*^2\sin\left(\frac{\beta_{32}^{1/2}\omega_0}{e_*}(t - t_{\mathrm{c}}(1-e_*))\right)\right],\end{aligned}\right\}\quad t_{\mathrm{c}} < t \leqslant t_{\mathrm{f}}. \tag{4.2 b}$$

For gross slip, the collision terminates and separation occurs at time $t_{\mathrm{f}}$, where $t_{\mathrm{f}} \equiv (1+e_*)t_{\mathrm{c}}$. This gives components of terminal relative displacement across the contact point

$$u_1(t_{\mathrm{f}}) = 0, \qquad u_3(t_{\mathrm{f}}) = (1 - e_*^2)u_{3\mathrm{c}}.$$

The contact force associated with these relative displacements is obtained from (2.2). Hence the components of contact force are obtained as

$$\left.\begin{aligned} F_1(t) &= -\mu\hat{s}F_3(t) = \mu\hat{s}\kappa u_{3\mathrm{c}}\sin(\beta_{32}^{1/2}\omega_0 t),\\ F_3(t) &= -\kappa u_{3\mathrm{c}}\sin(\beta_{32}^{1/2}\omega_0 t),\end{aligned}\right\}\quad 0 < t \leqslant t_{\mathrm{c}}\quad \text{(compression)}, \tag{4.3 a}$$

$$\left.\begin{aligned} F_1(t) &= e_*^2\mu\hat{s}\kappa u_{3\mathrm{c}}\sin\left[\frac{\beta_{32}^{1/2}\omega_0 t}{e_*}(t - t_{\mathrm{c}}(1-e_*))\right],\\ F_3(t) &= -e_*^2\kappa u_{3\mathrm{c}}\sin\left[\frac{\beta_{32}^{1/2}\omega_0 t}{e_*}(t - t_{\mathrm{c}}(1-e_*))\right],\end{aligned}\right\}\quad t_{\mathrm{c}} < t \leqslant t_{\mathrm{f}}\quad \text{(restitution)}. \tag{4.3 b}$$

The tangential impulse $p_k$ resulting from $F_k$ during the contact period $t_{\mathrm{f}}$ changes the relative velocity $v_k(t)$ across the contact point so that finally, at termination of contact, gross slip gives

$$\left.\begin{aligned} v_1(t_{\mathrm{f}}) &= v_1(0) + (1+e_*)\frac{\beta_2 + \mu\hat{s}\beta_1}{\beta_3 + \mu\hat{s}\beta_2}v_3(0),\\ v_3(t_{\mathrm{f}}) &= -e_* v_3(0).\end{aligned}\right\} \tag{4.4}$$

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### (*c*) *Limiting angle of incidence for gross slip*

At the contact point the sliding velocity $s(t)$ during compression can be expressed as

$$\begin{aligned} s(t) &= v_1(t) - \dot{u}_1(t) \\ &= v_1(0) + \frac{\beta_2 + \mu\hat{s}\beta_1}{\beta_3 + \mu\hat{s}\beta_2} v_3(0)[1 - \cos(\beta_{32}^{1/2}\omega_0 t)] + \mu\hat{s}\eta^2 v_3(0)\cos(\beta_{32}^{1/2}\omega_0 t). \end{aligned} \tag{4.5}$$

Thus initial sliding terminates during compression at time $t_\mathrm{s}$, where

$$\omega_0 t_\mathrm{s} = \beta_{32}^{-1/2} \arccos\left[\frac{v_1(0)/v_3(0) + (\beta_2 + \mu\hat{s}\beta_1)/(\beta_3 + \mu\hat{s}\beta_2)}{-\mu\hat{s}\eta^2 + (\beta_2 + \mu\hat{s}\beta_1)/(\beta_3 + \mu\hat{s}\beta_2)}\right] \tag{4.6}$$

if

$$\mu\eta^2 < -\frac{v_1(0)}{v_3(0)} < \frac{\beta_2 + \mu\hat{s}\beta_1}{\beta_3 + \mu\hat{s}\beta_2}.$$

Here the lower bound implies termination of sliding at time $t_\mathrm{s} = 0$ and the upper bound corresponds to compression terminating at time $t_\mathrm{s} = \omega_0^{-1}\pi/2$. In order for the upper bound to have a positive value, $\mu\hat{s}(0) > (\beta_3 - \beta_1)/(\beta_1 - \beta_2)$.

A condition for initial velocities that give continuous or *gross slip* during collision is obtained from the condition that the terminal tangential relative velocity vanishes at the instant of separation $t_\mathrm{f}$. This condition gives continuous sliding (in the initial direction) only if

$$\left|\frac{v_1(0)}{v_3(0)}\right| \geqslant (1 + e_*)\frac{\beta_2 + \mu\hat{s}\beta_1}{\beta_3 + \mu\hat{s}\beta_2}. \tag{4.7}$$

If $v_1(0) < 0$ and $\beta_2 > 0$, then initial sliding is slowed only if $(\beta_2/\beta_1) < \mu < (\beta_3/\beta_2)$.

## 5. Process of slip

### (*a*) *Small angle of incidence,* $|v_1(0)/v_3(0)| < \mu\eta^2$

If the initial tangential relative velocity is small, the contact initially sticks while the system undergoes a coupled two-degrees-of-freedom motion where the ratio of tangential to normal force is less than the coefficient of friction $\mu$. In systems where the ratio of tangential to normal stiffness $\eta^{-2} < 1$, the tangential compliant element is first compressed and then expands before the ratio of tangential to normal force first equals $\mu$. This equality occurs at a time $t_\sigma$ after the direction of the tangential force has reversed; it marks the beginning of a final period of slip. Figure 3 illustrates this process for a rigid slender bar that is inclined at an angle of inclination $\theta = \pi/6$ when it strikes a compliant, rough, inelastic half-space, $\mu = 0.5$, $e_* = 0.5$. At incidence the initial relative velocity at contact point C has both normal and tangential components. Small angles of incidence give initial stick–terminal slip behaviour. If $\beta_2 > 0$, the terminal period of slip is reverse slip, as illustrated, but if $\beta_2 < 0$, final slip is in the same direction as initial slip.

### (*b*) *Intermediate angle of incidence,* $\mu\eta^2 \leqslant |v_1(0)/v_3(0)| < (1 + e_*)[(\beta_2 + \mu\hat{s}\beta_1)/(\beta_3 + \mu\hat{s}\beta_2)]$

If the initial speed of sliding is somewhat larger, there is initial slip at the contact C in an initial period $t < t_\mathrm{s}$ during which the ratio of tangential to normal force is

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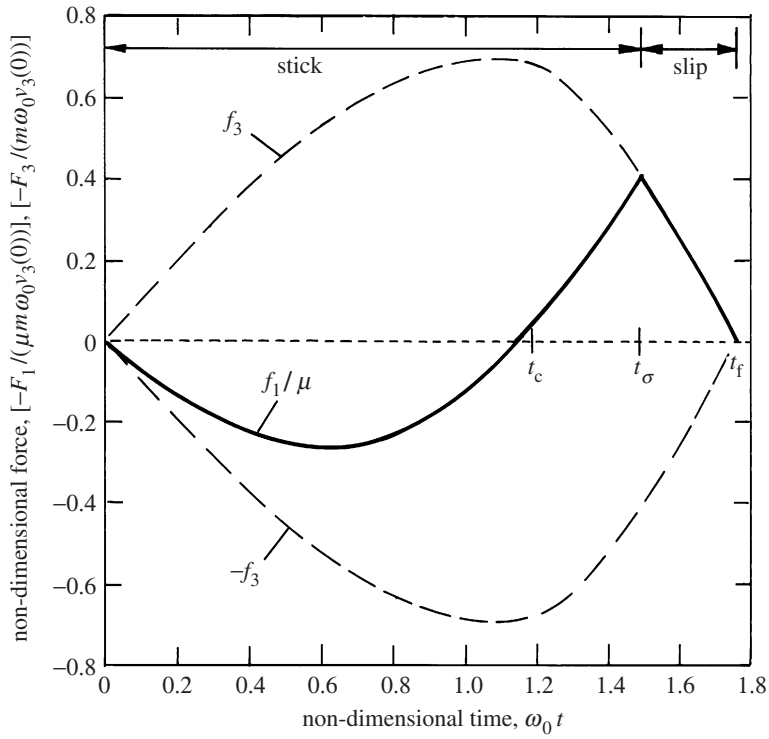


Figure 3. Normal and tangential components of contact force during impact between a rigid bar and a compliant, rough, inelastic half-space at a small angle of incidence, $|v_1(0)/v_3(0)| < \mu\eta^2$. Here the angle of inclination $\theta = \pi/6$, $e_* = 0.5$, $\mu = 0.5$ and $v_1(0)/v_3(0) = -0.4$.

$F_1(t)/F_3(t) = -\mu\hat{s}$. Initial sliding ceases and stick begins at time $t_s$ when first

$$\frac{dF_1(t_s)}{dt} \leqslant -\mu\hat{s}\frac{dF_3(t_s)}{dt}, \qquad t \leqslant t_s,$$
$$\frac{dF_1(t_s)}{dt} \geqslant -\mu\hat{s}\frac{dF_3(t_s)}{dt}, \qquad t > t_s,$$

where $dF_1/dt$ is obtained from equations for stick. This criterion determines the time $t_s$ when first the stick force obtained from the current initial state is less than that of limiting friction.

At time $t_s$ the speed of slip has slowed sufficiently so that stick can occur. In the following period of stick, the elastic element continues to reduce the initial relative tangential velocity until the direction of the tangential force reverses. This intermediate period of stick persists until the reversed tangential force is as large as the friction limit. At time $t_\sigma$ the intermediate period of stick transforms to a final period of sliding. During the terminal period of sliding the tangential force decreases because the normal compression of the contact region is decreasing until finally contact force vanishes at the time of separation $t_f$. This slip–stick–terminal slip behaviour is illustrated in figure 4 for $\beta_2 > 0$. Intermediate angles of incidence give slip–stick–reverse slip behaviour if $\theta > 0$ and $v_1(0) > 0$.

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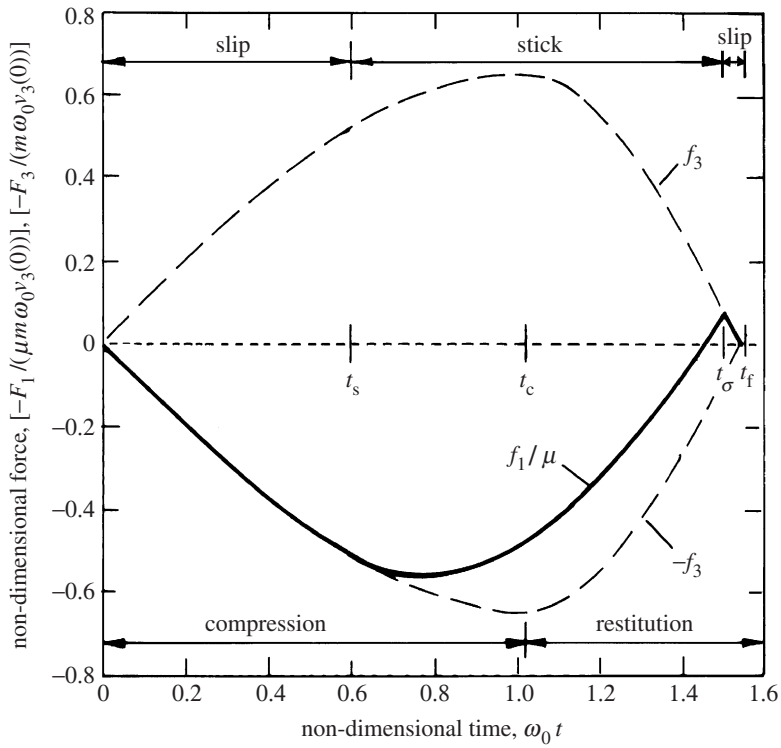


Figure 4. Normal and tangential components of contact force during impact between a rigid bar and a compliant, rough, inelastic half-space at an intermediate angle of incidence, $\mu\eta^2 \leqslant |v_1(0)/v_3(0)| < (1+e_*)(\beta_2+\mu\hat{s}\beta_1)/(\beta_3+\mu\hat{s}\beta_2)$. Here the angle of inclination $\theta = \pi/6$, $e_* = 0.5$, $\mu = 0.5$ and $v_1(0)/v_3(0) = -0.8$.

For frictional impact and an intermediate angle of incidence, figure 5 shows the contact force as a function of time for three different orientations of the colliding bar. Notice that for a negative angle of inclination $\theta < 0$, at the contact point there is slip–stick–slip without reversal of the direction of slip.

### (*c*) *Large angle of incidence,* $(1+e_*)[(\beta_2+\mu\hat{s}\beta_1)/(\beta_3+\mu\hat{s}\beta_2)] \leqslant |v_1(0)/v_3(0)|$

If the sliding speed is sufficiently large, there is continuous slip in the initial direction throughout the contact period. Gross slip occurs if the time of cessation of initial slip is as large as the time of separation, i.e. $t_s \geqslant t_f$. In this case the speed of slip only slows during impact; it does not vanish. Hence the change in relative velocity at the contact point is given by equation (4.4), and this velocity change is the same for kinematic (Newton), kinetic (Poisson) and energetic definitions of the coefficient of restitution (Stronge 2000).

Figure 6 illustrates the different slip processes which occur with increasing angle of incidence $\psi_0 = \tan^{-1}[-v_1(0)/v_3(0)]$; the normal and tangential forces are shown as functions of time for impact of a slender rigid bar inclined at an angle $\theta = \pi/6$ when one end strikes a compliant, rough, elastic half-space, $\mu = 0.5$, $e_* = 1$.

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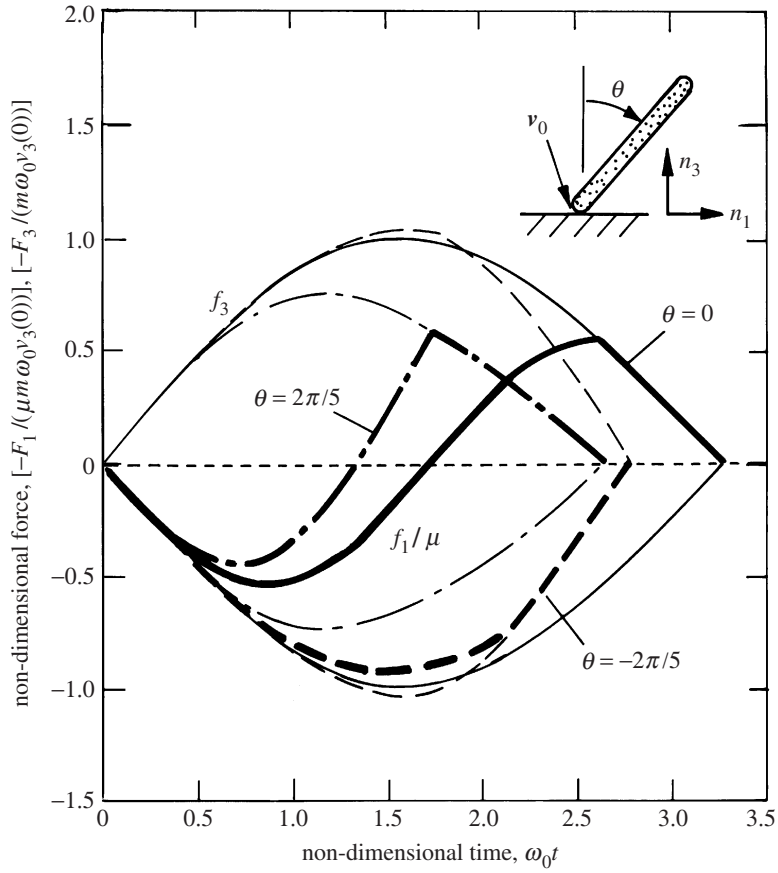


Figure 5. Variation of normal and tangential components of force for three different orientations of a bar striking a rough, inelastic half-space at an intermediate angle of incidence: $e_* = 1.0$, $\mu = 0.5$ and $v_1(0)/v_3(0) = -0.6$.

## 6. Discussion

### (a) *Example calculations*

In order to investigate effects of tangential compliance on collision configurations that are non-collinear, calculations have been performed mostly on a system consisting of a rigid uniform bar of mass $M$ inclined at an angle $\theta$ when one end strikes a rough inelastic half-space. The ratio of the half-length to the radius of gyration about the centre of mass for the rigid rod equals $\sqrt{3}$. For the system of a rigid bar striking a massive half-space, this gives inertia parameters

$$\beta_1 = 1 + 3(\cos\theta)^2,$$
$$\beta_2 = 3\sin\theta\cos\theta,$$
$$\beta_3 = 1 + 3(\sin\theta)^2.$$

The inelastic half-space has properties represented by a coefficient of limiting friction $\mu$, an energetic coefficient of restitution $e_*$, and elastic-plastic stiffness relations

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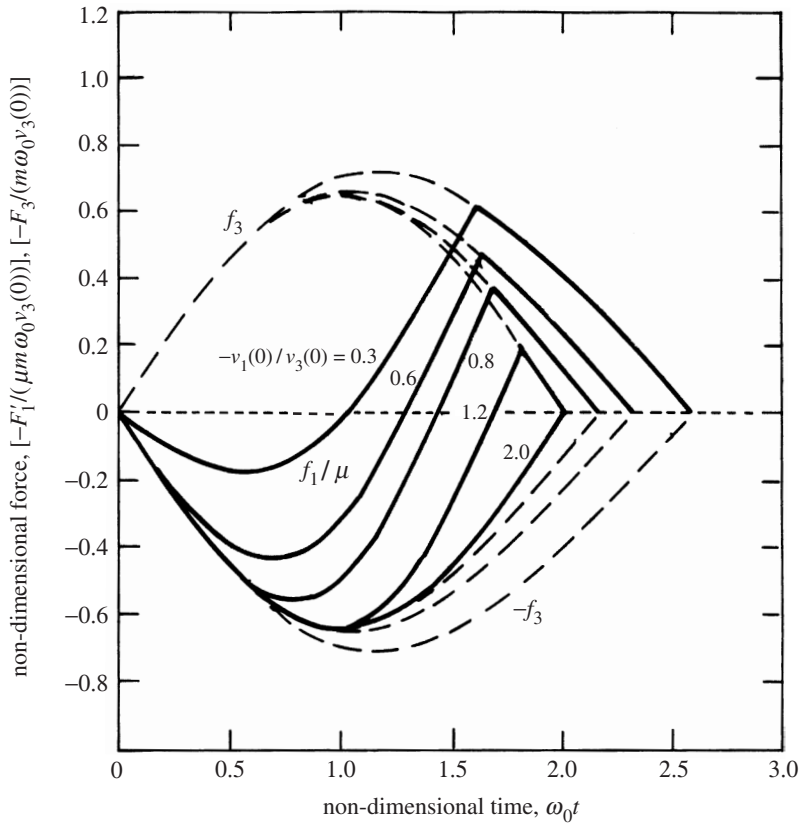


Figure 6. Variation of normal and tangential components of force on a bar striking a rough, inelastic half-space at small, intermediate and large angles of incidence: $\theta = \pi/6$, $e_* = 1.0$, $\mu = 0.5$ and $0.3 \leqslant -v_1(0)/v_3(0) \leqslant 2.0$.

as shown in figure 2. At incidence $t = 0$ the bar is assumed to be translating with uniform velocity

$$\boldsymbol{V}(0) = V_1(0)\boldsymbol{n}_1 + V_3(0)\boldsymbol{n}_3$$

and not rotating, $\dot{\boldsymbol{\theta}}(0) = 0\boldsymbol{n}_2$.

### (*b*) *Contact velocity at separation*

Experimental evidence for a reversal during collision in the direction of tangential relative velocity at the contact point C has been a principal factor pointing to the potential significance of tangential compliance. For collisions of spherical (collinear) bodies and three values of the coefficient of restitution, figure 7 illustrates the terminal tangential velocity as a function of the initial ratio of tangential to normal components of contact velocity, i.e. as a function of the angle of incidence. This graph shows reversal of the tangential velocity for small angles of incidence, but at large angles of incidence where there is gross slip, these curves become straight and there is no slip reversal. For comparison, if tangential compliance is negligible $\eta^2 \to 0$, the terminal tangential velocity vanishes to the left of the point where the curve crosses the line for no terminal slip.

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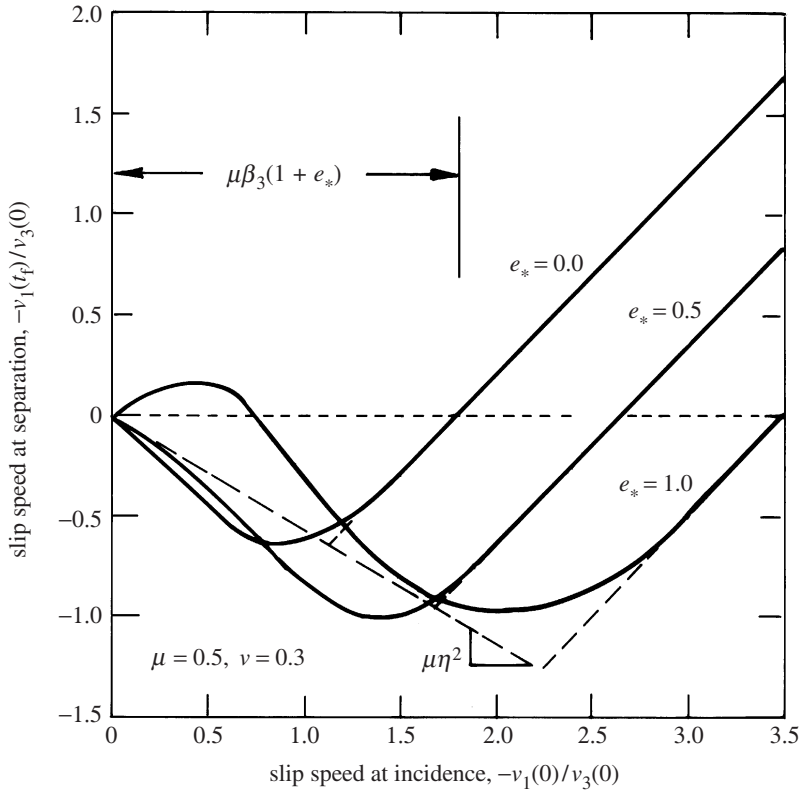


Figure 7. Variation of terminal slip speed $v_1(t_f)$ at contact point C with incident slip speed $v_1(0)$ for impact of a sphere against a rough, inelastic half-space and coefficients of restitution $e_* = 0.0$, 0.5 and 1.0: $\mu = 0.5$. Dashed lines are Walton's (1992) binary (stick or gross slip) approximation for collinear two-body contact. The coefficient for stick $\mu\eta^2$ was suggested by Foerster *et al.* (1994).

Similar curves for terminal tangential velocity at the colliding end of a rigid bar in a non-collinear configuration $\theta = \pi/3$ are shown in figure 8 for three different values of the coefficient of restitution. Even for negligibly small initial sliding, a non-collinear collision configuration results in tangential relative velocity. The figure shows that the angle of incidence for gross slip decreases as the coefficient of restitution $e_*$ decreases. This is in accord with the lower bound for gross slip (equation (4.7)).

Curves for terminal tangential velocity at the colliding end of a rigid bar as a function of the angle of incidence are shown in figure 9*a* for different orientations of the colliding bodies. The curves represent five angles of inclination for the bar, $\theta = -\pi/6, -\pi/12, 0, +\pi/12, +\pi/6$. Here the smallest angle of incidence which results in gross slip decreases with increasing inclination $\theta$ of the axis of the colliding bar from normal. At large and intermediate angles of incidence (and coefficient of restitution $e_* = 1$), these curves are identical to those of Jager (1997), who performed an elastic impact calculation based on an elastic continuum model of the traction distribution in the contact area during impact. For elastic bodies the discrete element and the elastic continuum models give slightly different results only in the range of small angles of incidence and small angles of inclination.

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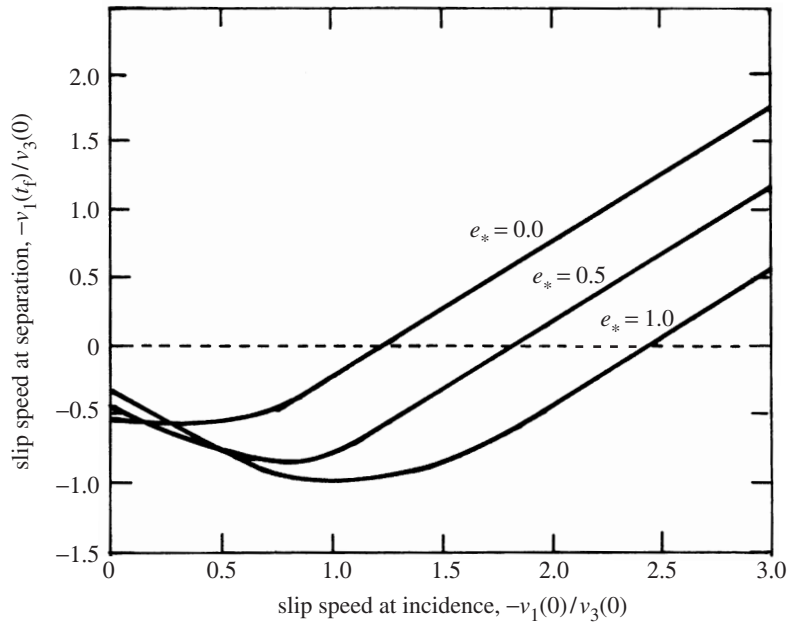


Figure 8. Variation of terminal slip speed $v_1(t_f)$ at contact point C with incident slip speed $v_1(0)$ for impact of a rigid bar against a rough, inelastic half-space and coefficients of restitution $e_* = 0.0$, 0.5 and 1.0: $\theta = \pi/6$ and $\mu = 0.5$.

### (*c*) *Effect of tangential compliance on change in slip speed*

The effect of tangential compliance on velocity changes during impact is illustrated clearly by these curves for the terminal tangential relative velocity as a function of the initial tangential relative velocity. For comparison with figure 9*a*, the curves in figure 9*b* are for a rigid bar striking a rough, elastic half-space that has negligible tangential compliance, $\eta^2 = 0$. In this example the coefficient of friction is larger than the coefficient for stick, $\bar{\mu}$, i.e. $\mu > \bar{\mu}$, where $\bar{\mu} \equiv \beta_2/\beta_1$. Consequently, for small or intermediate angles of incidence, initial sliding is brought to a halt before separation and subsequently the contact point sticks.† Comparison of parts (*a*) and (*b*) of figure 9 shows that the influence of tangential compliance is limited to impacts in the range of small or intermediate angles of incidence. If the initial slip speed is large enough to result in gross slip, tangential compliance is irrelevant.

### (*d*) *Angle of incidence for maximum friction at the contact point*

Ko (1985) performed experiments that imparted repeated impacts at oblique angles of incidence against a single point on a steel tube; these experiments showed that for small normal incident speeds, $v_3(0) < 1\ \mathrm{m\,s^{-1}}$, the wear rate due to fretting was closely correlated with the maximum force of friction, $F_{1\,\max}$. At large angles of incidence where there is gross sliding, the maximum friction force occurs simultaneously with the maximum normal force at termination of compression, $F_{1\,\max} =$

† Non-collinear collisions where the coefficient of friction is small, $\mu < \bar{\mu}$, and the angle of incidence is small or intermediate can result in slip reversal for orientations $\theta > 0$; alternatively, if $\theta < 0$, slip accelerates in the initial direction.

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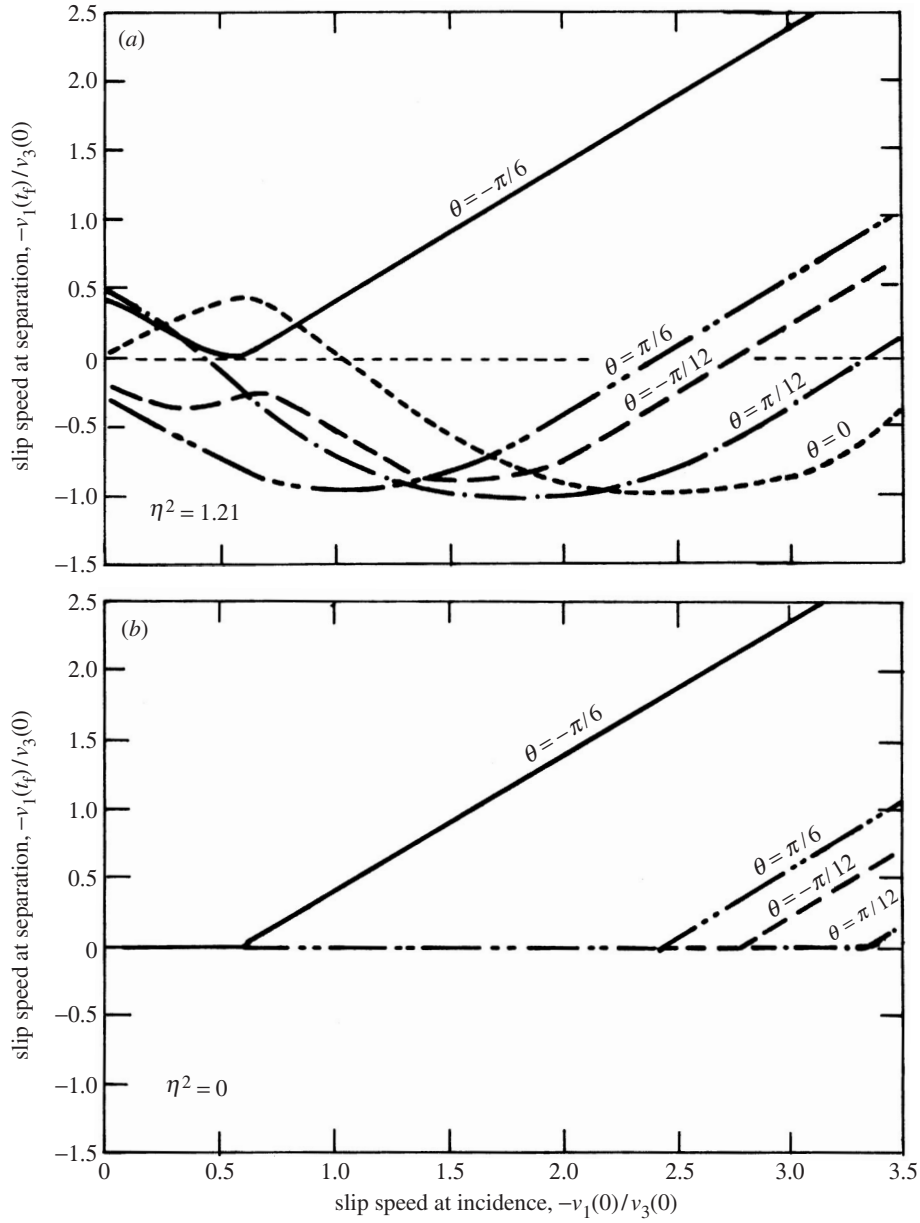


Figure 9. Variation of terminal slip speed $v_1(t_f)$ at contact point C with incident slip speed $v_1(0)$ for impact of a rigid bar against a rough, inelastic half-space and angles of inclination $\theta = -\pi/6, -\pi/12, 0, +\pi/12, +\pi/6$: $e_* = 1.0$ and $\mu = 0.5$. (*a*) $\eta^2 = 1.21$ and (*b*) $\eta^2 = 0$.

$-\mu\hat{s}F_3(\pi/2\omega_0)$. At small angles of incidence, however, the maximum tangential force occurs sometime during restitution, while, for intermediate angles of incidence, maximum friction occurs sometime during compression (see figures 3 and 4). Hence for small and intermediate angles of incidence, the maximum tangential force, $F_{1\,\max}$, depends on the angle of incidence, $\psi_0 = \tan^{-1}[-v_1(0)/v_3(0)]$.

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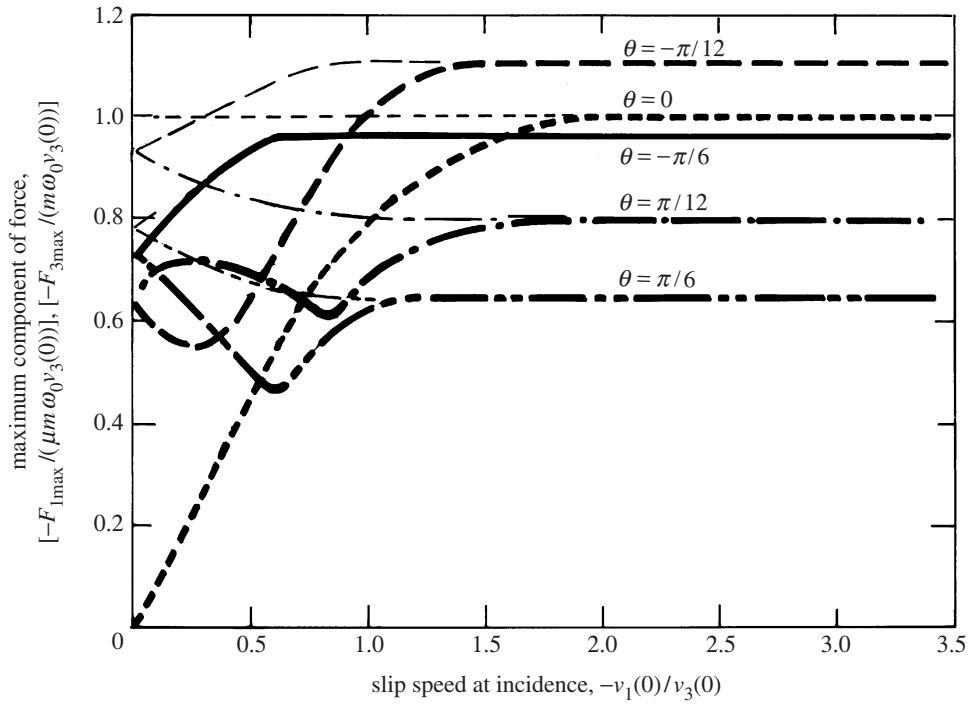


Figure 10. Maximum normal force and tangential force at contact point C as functions of incident slip speed $v_1(0)$ for impact of a rigid bar against a rough, inelastic half-space and angles of inclination $\theta = -\pi/6, -\pi/12, 0, +\pi/12, +\pi/6$: $e_* = 1.0$ and $\mu = 0.5$. Light lines for normal force $-F_{3\,\max}/m\omega_0 v_3(0)$ and heavy lines for tangential force $-F_{1\,\max}/\mu m\omega_0 v_3(0)$.

Figure 10 illustrates the variation of components of maximum contact force with the angle of incidence for five different orientations of a colliding body that has tangential compliance. In all cases the tangential relative velocity at incidence was positive. The curves represent maximum values of both the normal and an adjusted tangential component of contact force, $F_{3\,\max}$ and $F_{1\,\max}/\mu$. At large angles of incidence where there is gross slip, the normal and the adjusted tangential force coincide. For most impact configurations and angles of incidence, the tangential force resulting from gross slip is larger than that obtained from small or intermediate angles of obliquity.

### (*e*) *Terminal ratio of tangential to normal impulse*

To examine the hypothesis that effects of friction on velocity changes during impact can be calculated with use of an experimentally measured ratio of tangential to normal terminal impulse, the ratio of impulses $p_1(t_\mathrm{f})/p_3(t_\mathrm{f})$ has been calculated for a rigid bar striking a rough, inelastic half-space. Figure 11 shows this ratio as a function of the angle of incidence at the contact point for bars inclined at five different angles. In the range of small to intermediate angles of incidence, the terminal ratio of tangential to normal impulse is a complicated function of the impact configuration as well as the angle of incidence and the coefficient of restitution.

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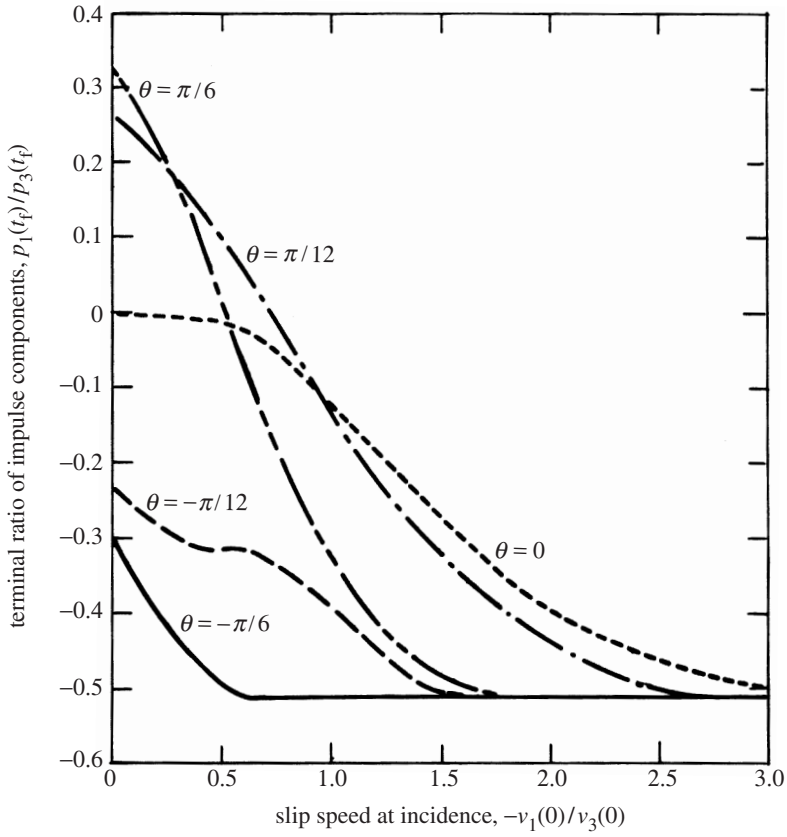


Figure 11. Terminal ratio of tangential to normal impulse $p_1(t_f)/p_3(t_f)$ for impact of a rigid bar against a rough, inelastic half-space when the bar is oriented at angle of inclination $\theta = -\pi/6, -\pi/12, 0, +\pi/12, +\pi/6$. Impact parameters $e_* = 1.0$ and $\mu = 0.5$.

## 7. Conclusions

For a non-collinear collision between a pair of hard, inelastic bodies, a discrete-element model of local contact compliance has been analysed for effects due to tangential compliance. For planar impact this approach results in coupled differential equations for normal and tangential relative accelerations at the contact point, i.e. the analysis considers impact as a smooth dynamic process in order to calculate small, time-dependent relative displacements in the contact region. As noted previously by García-Reyero Viñas (2000), the solution is independent of the normal compliance $\kappa^{-1}$ at the contact point; it depends on the ratio of tangential to normal compliance $\eta^2$ but not on the magnitude of the compliance.

For planar collisions with friction, the slip process at the contact point is either stick–slip, slip–stick–slip or continuous slip; these processes result from small, intermediate or large angles of incidence, respectively. Surprisingly, the stick–slip process occurs within a range of angles of incidence that is independent of eccentricity of the colliding bodies, i.e. the largest angle of incidence for initial stick is independent of the impact configuration. On the other hand, the smallest angle of incidence for

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gross slip is dependent on the impact configuration; the lower limit angle of incidence for gross slip increases as the eccentricity of the impact configuration increases.

The effect of elastic tangential compliance on the outcome of a collision is limited to angles of incidence in the small and intermediate range, i.e. tangential compliance has no effect on the terminal state of the system if the angle of incidence is large so that there is gross slip. For small or intermediate angles of incidence, tangential compliance results in reversal of the direction of slip unless the angle of inclination is such that $\beta_2$ is negative and relatively large. For small or intermediate angles of incidence, tangential compliance significantly affects the change of state during impact; its effect is as large as that of friction.

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